

AEL HW8

1) a) Gain Margin: find f_{-180} from: $\angle T(jf_{-180}) = -180$
and then calculate $|T(jf_{-180})|$ to get

$$GM = -20 \log_{10} |T(jf_{-180})|$$

$$\angle T(jf_{-180}) = -\underbrace{\arctan\left(\frac{f_{-180}}{5}\right)} - \arctan\left(\frac{f_{-180}}{155}\right) - \arctan\left(\frac{f_{-180}}{10^6}\right) = -180^\circ$$

this one could be assumed to be -90°
since $f_{-180} \gg 5 \text{ Hz}$

$$f_{-180} = 316 \text{ kHz (calculator)}$$

substituting & solving $|T(jf_{-180})|$ gives:

$$|T(jf_{-180})| = 0.4545 = -6.849 \text{ dB}$$

$$\rightarrow GM = 6.849 \text{ dB}$$

b) Phase margin: find f_x from $|T(jf_x)| = 1$ and then
calculate $\angle T(jf_x)$ to get $\phi_m = 180 - \angle T(jf_x)$

$$|T(jf_x)| = 1 = 10^5 / \sqrt{\left(1 + \frac{f_x^2}{5^2}\right) \left(1 + \frac{f_x^2}{10^{10}}\right) \left(1 + \frac{f_x^2}{10^{12}}\right)}$$

$$f_x = 210 \text{ kHz (calculator)}$$

substituting & solving $\angle T(jf_x)$ gives:

$$\angle T(jf_x) = -166.4^\circ$$

$$\phi_m = 13.6^\circ$$

c) find f_x such that $\phi_m = 45^\circ : \rightarrow \angle T(jf_x) = -135^\circ$

$$\angle T(jf_x) = -135^\circ = -\arctan(f/5) - \arctan(f/10^5) - \arctan(f/10^6)$$

$$\rightarrow f_x = 84.4 \text{ kHz}$$

want $|T(jf_x)| = 1$ with a changed β , assume $\beta' = x\beta$ then

$$|T(jf_x)| = 1 = |a(jf_x)\beta'| = x |a(jf_x)\beta(jf_x)| = x |T(jf_x)|$$

$$x = \frac{1}{4.5}, \text{ therefore, } \beta \text{ must be decreased } 1/4.5 \text{ or to } 22\%$$

d) Do the same as above, but with $\phi_m = 0^\circ, \angle T = -180$

or set $GM = 0$ and solve the other way.

both will result in same answer because f_{180} will equal f_x

$\angle T$ will not change with β scaling, so take f_{180} from (a) and set equal to f_x ,

$$f_x = 316 \text{ kHz}$$

we also already know $|T(jf_{180})| = 0.454 = |T(jf_x)|$

to get to precipice of instability, do same as above:

$$|T'(jf_x)| = 1 = x |T(jf_x)| \rightarrow x = \frac{1}{0.454} = 2.2$$

therefore, increase be to just less than $2.2x$ or 220%

2 | from description: $a(jf) = \frac{2 \times 10^5}{(1+jf/f_1)(1+jf/f_2)}$

where $f_1 = 100 \text{ kHz}$, $f_2 = 50 \text{ MHz}$

a) assuming β is resistive, then simplified two-pole quality factor equations can be used. Knowing GP occurs when $Q > 1/\sqrt{2}$, then assume $Q = 1/\sqrt{2}$

$$1/\sqrt{2} = f_0 / (f_1 + f_2) \rightarrow f_0 = \frac{100\text{k} + 50\text{M}}{\sqrt{2}} = 35.4 \text{ MHz}$$

$$f_0^2 = a_0 \beta f_1 f_2 = (2 \times 10^5) \beta (100\text{k})(50\text{M})$$

$$\beta \leq 0.00126$$

Non-inverting amp: we know $1/\beta = A_{\text{ideal}}$
therefore:

$$A_{\text{ideal}} \geq 1/\beta = 796.8$$

2b Add a low-freq pole such that amp has $\phi_m = 60^\circ$ with $\beta = 1$

$$a(jf) = \frac{2 \times 10^5}{(1+jf/f_3)(1+jf/f_1)(1+jf/f_2)} \quad f_3 \rightarrow \text{low freq}$$

since $\beta = 1$, then $T(jf) = a(jf)$

$$\phi_m = 60^\circ = 180^\circ + \angle T(jf) \\ = 180^\circ - \underbrace{\text{atan}(f/f_3)} - \text{atan}(f/f_1) - \text{atan}(f/f_2)$$

this is low-freq, therefore,
we can assume that at higher f , $\angle = -90^\circ$

$$60^\circ \approx 90^\circ - \text{atan}(f/f_1) - \text{atan}(f/f_2)$$

could solve this numerically to get f .

But! we know that f must be near f_1 to get $\phi_m = 60^\circ$, therefore, $f \ll f_2$ and therefore the phase contribution of $f_2 \approx 0^\circ$

$$60^\circ \approx 90^\circ - \text{atan}(f/f_1)$$

$$\rightarrow f = 57.7 \text{ kHz}$$

Need $|T| \leq 1$ at this frequency:

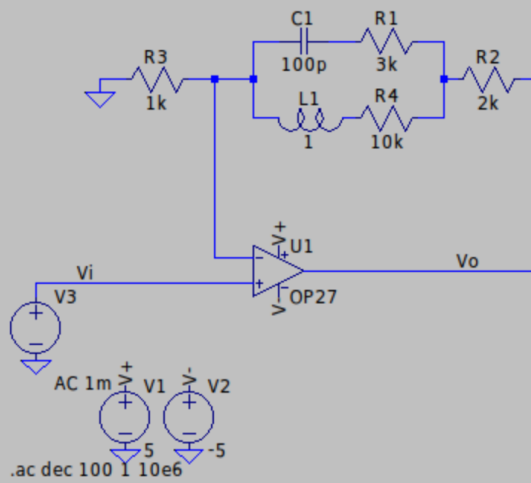
$$1 \geq 2 \times 10^5 / |1+jf/f_3| |1+jf/100k| |1+jf/500k|$$

solve for $f_3 \dots$

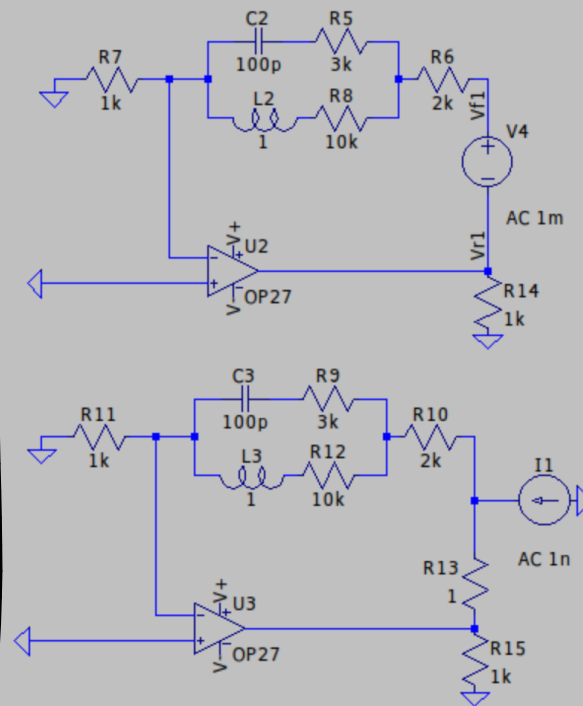
$$\sqrt{1+f^2/f_3^2} \geq 1.73 \times 10^5 \rightarrow f_3 \leq 0.33 \text{ Hz}$$

3) Circuits

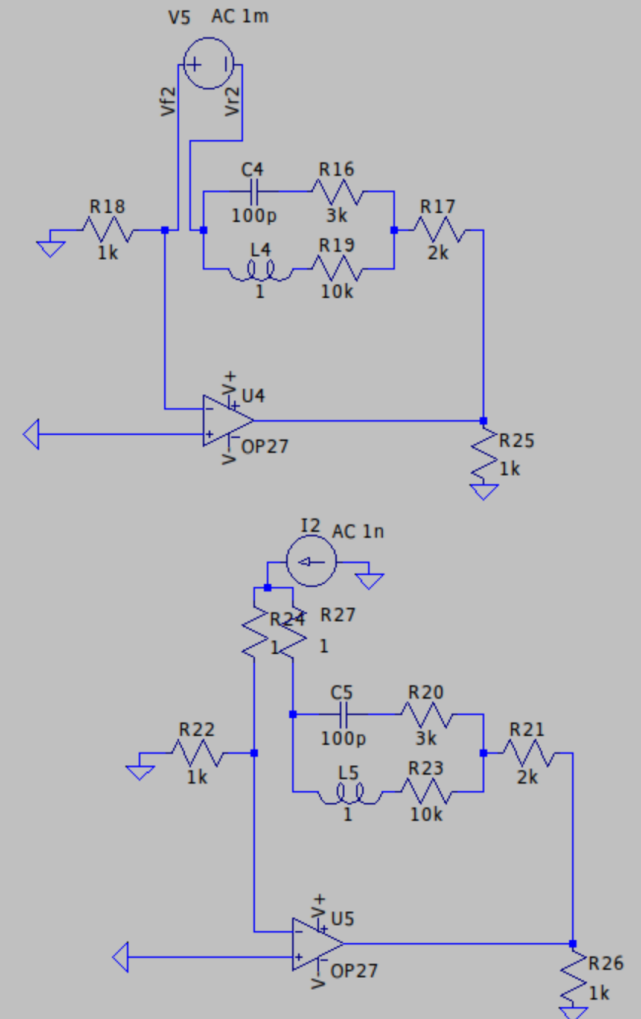
Base Circuit (Unused)



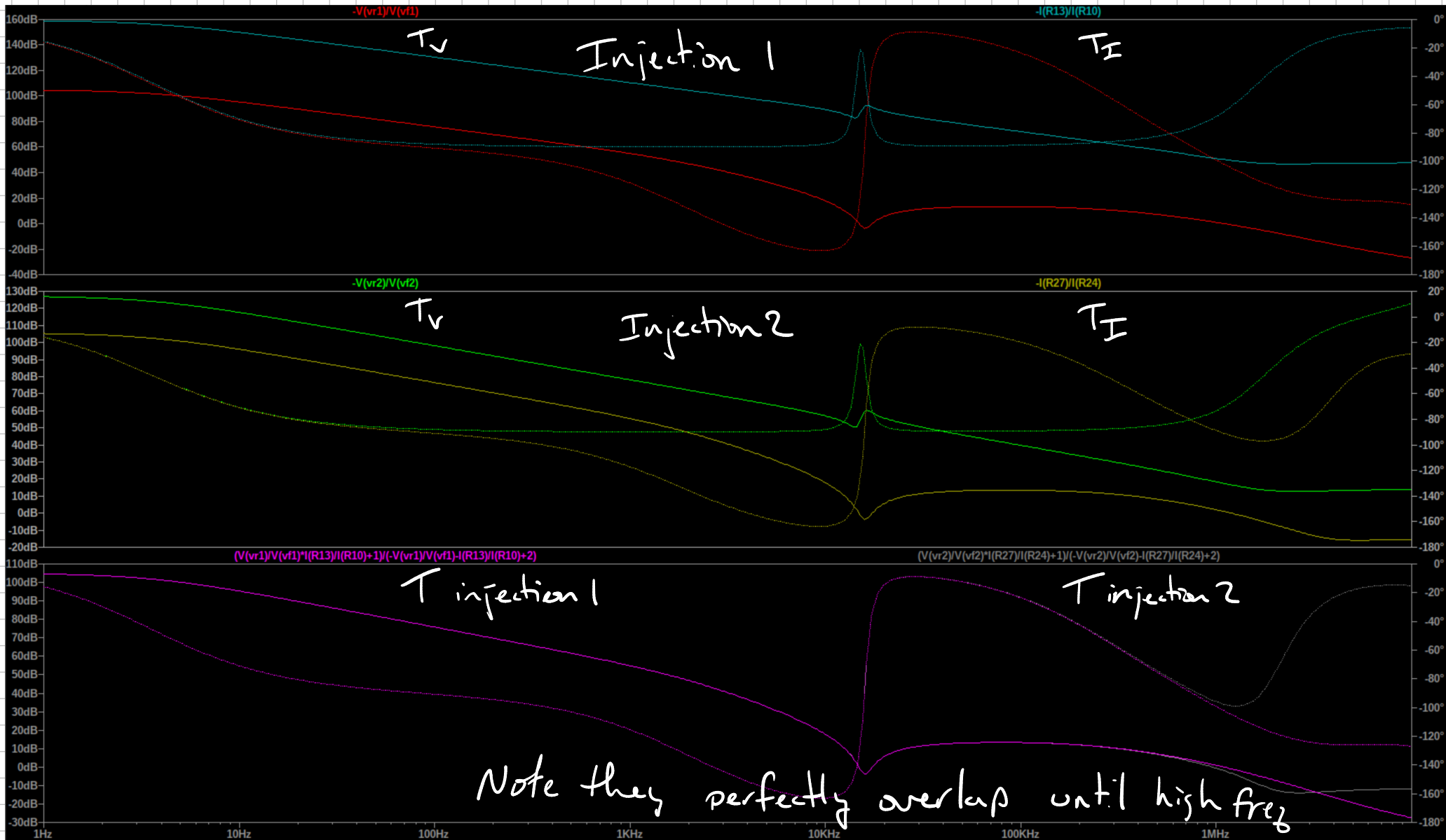
Injection 1



Injection 2



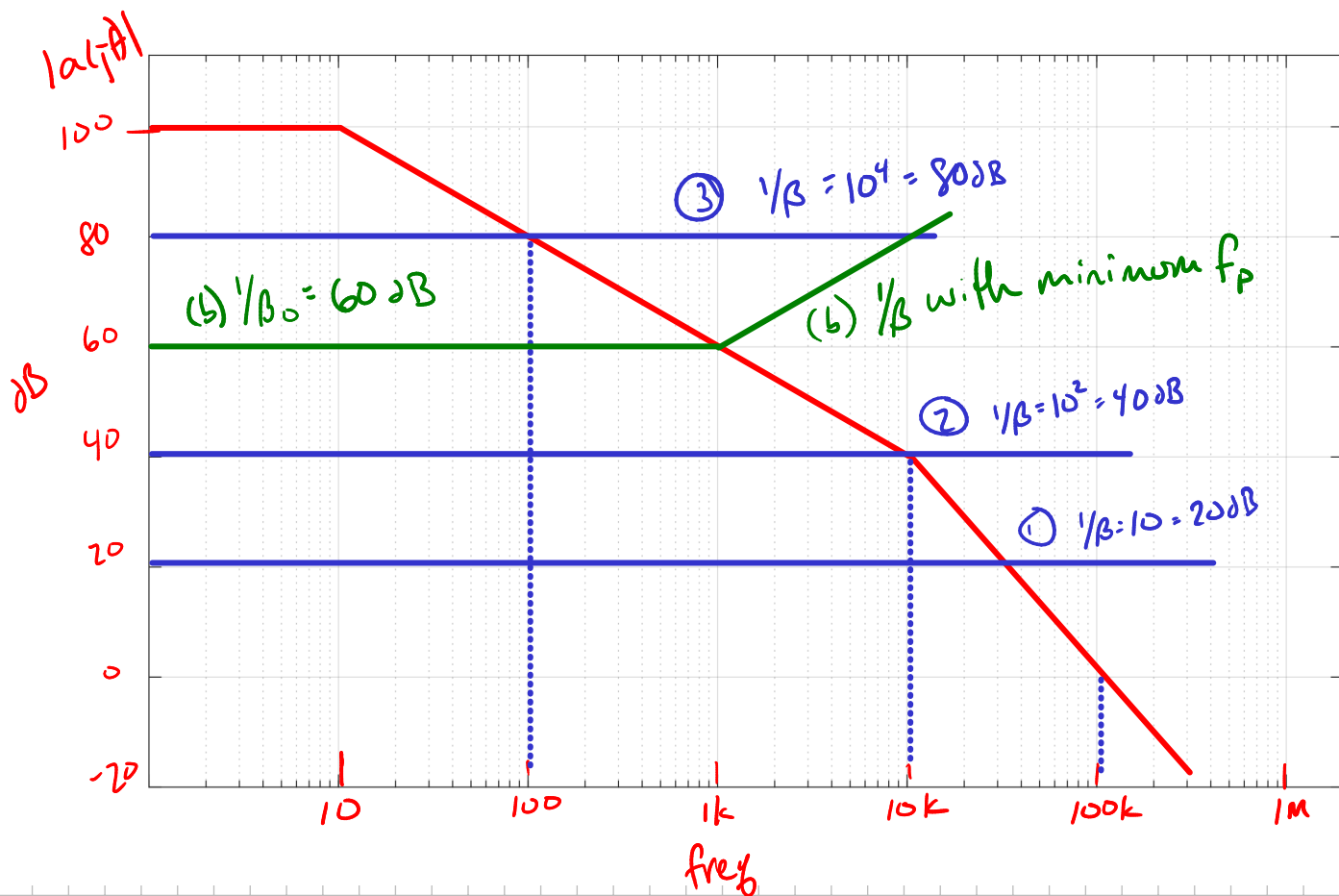
R13,R24,R27 added for current measurement (small enough to not affect circuit)



Assume these are due to simulation error

$$4) a(j\omega) = 10^5 / (1 + jf/10)(1 + jf/10^4)$$

$$20 \log_{10} 10^5 = 100 \text{ dB}$$



$a) 1/\beta:$	$f_x:$	ROL	ϕ_m
① $1/0.1 \rightarrow 20 \text{ dB}$	100 kHz	$0 - 40 = 40$	0°
② $1/10^2 \rightarrow 40 \text{ dB}$	10 kHz	$0 - 30 = 30$	45°
③ $1/10^4 \rightarrow 80 \text{ dB}$	100 Hz	$0 - 20 = 20$	90°

$$b) A = 1/10^3 (1 + jf/f_p) \rightarrow 1/\beta = 10^3 (1 + jf/f_p) \quad \beta_0 = \beta(j\omega)|_{f \rightarrow 0} = 1/10^3$$

$$1/\beta_0 = 10^3 = 60 \text{ dB}$$

minimum f_p occurs @ $f_x \rightarrow f_p \geq 1 \text{ kHz}$